Two positive Fisher–KPP equilibria on a uniformly smooth unbounded domain

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Abstract

We construct a connected unbounded domain in $\mathbb{R}^{21}$ with uniformly $C^{2,1/2}$ boundary on which the Dirichlet Fisher–KPP equation has two distinct bounded positive equilibria. The boundary consists of two smooth global graphs whose separation tends to the critical slab width. A corrected transverse mode gives a slowly decaying subsolution, while a compact enlargement of the domain permits a second equilibrium with a faster upper bound. The example gives a negative answer to the unrestricted strong-KPP uniqueness question. No optimal dimension is asserted.

1 The uniqueness question

Positive stationary populations with diffusion and density-dependent growth are described by

$$-\Delta u = f(u) \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{1}$$

Here Ω is the habitat and the boundary is absorbing. A decreasing per-capita growth rate, meaning that $s \mapsto f(s)/s$ is strictly decreasing on $(0, 1]$, is called the strong-KPP condition here. It gives uniqueness of a positive equilibrium on smooth bounded domains. On unbounded domains, positive solutions may approach zero at infinity, and a global comparison between two solutions is harder to obtain.

Berestycki and Graham proved uniqueness on general uniformly smooth domains under a non-degeneracy condition on the principal limit spectrum [1, Theorem 1.1], and asked whether the condition could be removed [1, Conjecture 1.3]. Their later paper retains the general question [2, Section 8.1, Open Question 3]. Recently, Rossi and Tough proved uniqueness for the quadratic Fisher–KPP reaction through dimension 6, and constructed homogeneous counterexamples on locally analytic domains in every dimension at least 7 [3, Theorems 1.5, 1.8 and 7.1]. Their homogeneous construction uses spherical obstacles with radii tending to zero. The example below instead has two global boundary graphs with uniform derivative bounds.

To keep both the reaction and its derivative bounded on $[0, \infty)$ while retaining the logistic reaction on $[0, 1]$, set

$$f(s) = \begin{cases} s - s^2, & 0 \le s \le 1, \\ s^{-1} - 1, & s \ge 1. \end{cases} \tag{2}$$

This function is bounded with bounded Lipschitz derivative. In particular, $f \in C^{1,1/2}([0, \infty))$, $f(0) = f(1) = 0$, $f'(0) = 1$, $f > 0$ on $(0, 1)$, $f < 0$ on $(1, \infty)$, and $f(s)/s = 1 - s$ strictly decreases on $(0, 1]$.

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Theorem 1. *There is a connected unbounded domain $\Omega \subset \mathbb{R}^{21}$ with uniformly $C^{2,1/2}$ boundary for which (1), with (2), has two distinct solutions $U, T \in C^{2,1/2}_{\mathrm{loc}}(\overline{\Omega})$ satisfying $0 < U, T \le 1$ in Ω. The same functions solve $-\Delta u = u - u^2$ with zero Dirichlet data.*

The Dirichlet condition is the admissible value $\beta = 0$ of $\beta\partial_n u + (1-\beta)u = 0$. Thus one example in Theorem 1 refutes the universal uniqueness assertion for all allowed dimensions, domains, reactions and boundary parameters. It makes no classification claim for other dimensions or for Robin boundary conditions.

The domain is a strip whose width approaches the critical value π with an inverse-square excess, together with a compact central enlargement. We first construct an equilibrium bounded below by a multiple of $|x|^{-2}$. The enlargement also supports a compact positive subsolution. An exterior supersolution of order $|x|^{-4}$ can be joined to the first equilibrium and used to construct the second one. The different bounds at infinity separate the two equilibria.

2 Comparison functions on an unbounded strip

We use the polynomial residual $\mathcal{F}(v) = \Delta v + v - v^2$. A continuous, locally H^1 function is a subsolution if $\mathcal{F}(v) \ge 0$ in distributions, and a supersolution if $\mathcal{F}(v) \le 0$. All subsolutions and supersolutions used to construct equilibria lie in $[0, 1]$, where the reaction in (2) agrees with the polynomial. The following existence fact records how the boundary at infinity is treated.

Lemma 2. *Let $\ell : \mathbb{R}^{20} \to (0, \infty)$ be smooth, bounded above and below by positive constants, with bounded derivatives, and put $D = \{(x, y) : |y| < \ell(x)\}$. Suppose a, c are continuous and locally H^1 up to ∂D, $0 \le a \le c \le 1$, $\mathcal{F}(a) \ge 0$, $\mathcal{F}(c) \le 0$, and $a = 0$ on ∂D. Then there is a classical solution u of $\Delta u + u - u^2 = 0$ on D, with $a \le u \le c$ and $u = 0$ on ∂D. It is locally $C^{2,1/2}$ up to the boundary. If $a \not\equiv 0$, then $u > 0$.*

Proof. Let $D_n = D \cap \{|x| < n\}$. This is a bounded Lipschitz domain: its end face meets the two graph faces transversely. Prescribe the boundary extension a, so that the trace is zero on the true boundary and a on the end face. Set $u_0 = a$ and solve

$$(-\Delta + 1)u_{j+1} = G(u_j), \qquad G(t) = 2t - t^2, \qquad u_{j+1} - a \in H^1_0(D_n). \tag{3}$$

The weak linear Dirichlet theorem [4, Theorem 8.9] applies with constant principal coefficients and zero-order coefficient -1 after reversing signs. Testing a difference with its positive part proves the comparison principle for $-\Delta + 1$. Since G increases on $[0, 1]$, the subsolution and supersolution inequalities give $a = u_0 \le u_1 \le \cdots \le c$.

For fixed n, let v be the almost everywhere monotone limit. The bounded functions $G(u_j)$ converge to $G(v)$ in $L^2(D_n)$. Let u_* solve the linear problem with this limiting right hand side and the same boundary extension. The energy identity for the difference gives

$$\|u_{j+1} - u_*\|_{H^1(D_n)} \le \|G(u_j) - G(v)\|_{L^2(D_n)} \longrightarrow 0.$$

Comparison with a, c identifies $u_* = v$ and gives a weak semilinear solution u_n between the two functions.

On every fixed interior patch, or fixed true-boundary patch away from the end face, $0 \le u_n \le 1$ and $\Delta u_n = -u_n + u_n^2$ has bounded right hand side. Local weak boundary regularity [4, Corollary 8.36], with zero boundary data and constant coefficients, gives uniform local $C^{1,1/2}$ bounds. The right hand side is therefore uniformly $C^{0,1/2}$ on smaller patches. Theorem 8.9 also gives local

H^2 regularity; Theorem 9.19, with $k = 1$, then gives interior $C^{2,1/2}$ regularity. Lemma 6.18 and the local estimates of Corollaries 6.3 and 6.7 in [4] give uniform $C^{2,1/2}$ bounds on smaller true-boundary and interior patches. All these constants concern fixed patches; the distant end face does not enter them.

A diagonal subsequence converges locally in C^2, including compact portions of the true boundary. Its limit satisfies the equation, ordering and zero boundary condition. If a is nonzero, the limit is nonzero. Since $\Delta u = -u(1-u) \le 0$, the strong minimum principle on the connected domain gives $u > 0$. No global energy norm or prescribed decay at infinity is used. □

Two elementary facts will also be useful. If a smooth subsolution v vanishes on the boundary of a subdomain E and $\partial_n v \le 0$, then its zero extension $\widetilde{v}$ is a weak subsolution. Indeed, for every nonnegative compactly supported smooth test function ζ,

$$\int (\widetilde{v}\Delta\zeta + (\widetilde{v} - \widetilde{v}^2)\zeta) = \int_E \mathcal{F}(v)\zeta - \int_{\partial E} \partial_n v\, \zeta \ge 0. \tag{4}$$

The formula holds face by face for a cube. The extensions are locally H^1 because their traces agree with zero.

The minimum of two smooth supersolutions is a weak supersolution. To see this without a monotonicity assumption on the reaction, approximate the minimum of v, w by $m_\varepsilon = (v + w - \sqrt{(v-w)^2 + \varepsilon^2})/2$. Differentiation gives $\Delta m_\varepsilon \le \theta_\varepsilon \Delta v + (1-\theta_\varepsilon)\Delta w$, where $0 \le \theta_\varepsilon \le 1$. Thus $\mathcal{F}(m_\varepsilon)$ is bounded above by

$$(m_\varepsilon - m_\varepsilon^2) - \theta_\varepsilon(v - v^2) - (1-\theta_\varepsilon)(w - w^2).$$

This tends to zero pointwise and is bounded on every compact set, including where $v = w$. Testing and passing to the limit proves the assertion. The minimum is locally H^1 by the weak chain rule [4, Section 7.4 and pp. 153–154].

3 Two different rates of decay

Write $x \in \mathbb{R}^{20}$, $y \in \mathbb{R}$, and fix

$$R = 10^6, \quad b = 20000, \quad q = R^2 + |x|^2, \quad k = 1 - \frac{24}{q}, \quad \ell_0 = \frac{\pi}{2k}, \quad \phi(x,y) = \cos(k(x)y). \tag{5}$$

Let $\Omega_0 = \{(x,y) : |y| < \ell_0(x)\}$. Then $q \ge 10^{12}$, $1/2 \le k < 1$, $|y| < \pi$ and $0 < \phi \le 1$ in Ω_0. The transverse mode ϕ vanishes on its boundary. We need to retain the derivatives of this mode in x, since their signs matter near the boundary.

Lemma 3. *On Ω_0, the functions*

$$S = q^{-1}(\phi + bq^{-2}\phi^2), \qquad V_B = Bq^{-2}\phi \quad (B > 0)$$

satisfy $0 < S \le 1$, $\mathcal{F}(S) \ge 0$ and $\mathcal{F}(V_B) \le 0$. They vanish on $\partial\Omega_0$, and $\partial_n S < 0$ there.

Proof. For a positive amplitude $H = H(x)$, direct differentiation gives

$$\begin{aligned}\frac{\mathcal{F}(H\phi)}{H} = \phi\left(\frac{\Delta_x H}{H} + 1 - k^2 - y^2|\nabla k|^2 - H\phi\right) \\ - y\sin(ky)\left(\Delta_x k + 2\nabla \log H \cdot \nabla k\right).\end{aligned} \tag{6}$$

For $H = Aq^{-p/2}$ and $r = |x|$,

$$\frac{\Delta_x H}{H} = -\frac{20p}{q} + \frac{p(p+2)r^2}{q^2},$$
$$\nabla k = \frac{48x}{q^2}, \qquad \Delta_x k = \frac{960}{q^2} - \frac{192r^2}{q^3},$$
$$\Delta_x k + 2\nabla \log H \cdot \nabla k = \frac{960}{q^2} - \frac{96(p+2)r^2}{q^3}. \tag{7}$$

For $p = 4$, the first linear coefficient in (6) is at most $(-56+48)/q = -8/q$, while the last bracket is at least $384/q^2$. Both ϕ and $y\sin(ky)$ are nonnegative. The other terms have nonpositive sign, proving $\mathcal{F}(V_B) \le 0$ for every amplitude $B > 0$.

For the slow subsolution put $h = q^{-1}$ and $\delta = bq^{-2}$. The last bracket in (6) lies between $576/q^2$ and $960/q^2$. Using $\pi < 4$ and $|\nabla k|^2 \le 2304q^{-3}$ gives

$$\mathcal{F}(h\phi) \ge h\phi\left(\frac{7}{q} - \frac{576}{q^2} - \frac{36864}{q^3}\right) - \frac{3840h}{q^2} \ge \frac{6h\phi}{q} - \frac{3840h}{q^2}. \tag{8}$$

The negative term need not contain ϕ, which is why the squared mode is added. With $g = \delta h = bq^{-3}$, its exact contribution is

$$\mathcal{F}(S) = \mathcal{F}(h\phi) + g\big[2k^2 + (1-4k^2)\phi^2\big] + E, \quad E = \Delta_x(g\phi^2) - 2\delta h^2\phi^3 - \delta^2 h^2\phi^4. \tag{9}$$

To bound the remainder uniformly, (7) gives

$$|\nabla_x \phi| \le 192q^{-3/2}, \quad |\Delta_x \phi| \le 40704q^{-2}, \quad |\nabla_x(\phi^2)| \le 384q^{-3/2}, \quad |\Delta_x(\phi^2)| \le 155136q^{-2}.$$

Also $|\nabla g| \le 6bq^{-7/2}$ and $|\Delta_x g| \le 168bq^{-4}$. The product rule and the two nonlinear terms in (9) yield

$$|E| \le (159914b + b^2)q^{-4} < 4 \cdot 10^9 \frac{h}{q^3}. \tag{10}$$

For $\phi \le 1/4$, the square bracket in (9) is at least $5/16$, and hence

$$\mathcal{F}(S) \ge \left(6250 - 3840 - \frac{4 \cdot 10^9}{q}\right)\frac{h}{q^2} > 0.$$

For $\phi \ge 1/4$ the bracket is at least -3, giving

$$\mathcal{F}(S) \ge \left(\frac{3}{2} - \frac{63840}{q} - \frac{4 \cdot 10^9}{q^2}\right)\frac{h}{q} > 0.$$

Both inequalities hold for all $q \ge 10^{12}$. Finally $S \le q^{-1}(1 + bq^{-2}) < 1$. On the boundary its normal derivative is $h\partial_n \phi = -h|\nabla\phi| < 0$, since $|\partial_y \phi| = k$ there. □

4 A compact enlargement and two equilibria

Choose a smooth radial cutoff $0 \le \chi \le 1$ which is 1 for $r \le 2000$ and 0 for $r \ge 4000$, and set

$$\ell(x) = \ell_0(x) + 2000\chi(x), \qquad \Omega = \{(x,y) : |y| < \ell(x)\}. \tag{11}$$

For example, if $\eta(t) = e^{-1/t}$ for $t > 0$ and $\eta(t) = 0$ otherwise, one can take $\chi = \eta(4000^2 - r^2)/[\eta(4000^2 - r^2) + \eta(r^2 - 2000^2)]$. Its denominator is positive. The two boundary graphs have bounded derivatives through order 3 and have separation at least π. They therefore admit common-radius $C^{2,1/2}$ charts with uniformly bounded norms. The domain is connected through its central plane $y = 0$, is unbounded, and agrees with Ω_0 for $r \geq 4000$.

Extend S by zero from Ω_0 into Ω. By (4) and Lemma 3, it is a weak subsolution. It lies below the constant supersolution 1. Lemma 2 gives an equilibrium U on Ω with

$$0 < U \leq 1, \qquad U(x, 0) \geq q^{-1}. \tag{12}$$

We next construct a supersolution whose exterior bound is faster. On the compact collar $4500 \leq r \leq 5500$, $|y| \leq \ell_0(x)$, let $M = \sup |\partial_y U| < \infty$. For $d = \ell_0(x) - |y|$, integration from the nearest endpoint gives $U \leq Md$. On the other hand,

$$\phi = \sin(kd) \geq \frac{2kd}{\pi} \geq \frac{d}{\pi}.$$

The finite choice $B = 1 + \pi M(R^2 + 5500^2)^2$ consequently gives $V_B \geq U$ on this collar. Define

$$W = \begin{cases} U, & r \leq 5000, \\ \min(U, V_B), & r > 5000. \end{cases} \tag{13}$$

It equals U on a neighborhood of $r = 5000$. The minimum property proved above and Lemma 3 show that W is a weak supersolution on the whole of Ω. Moreover $0 < W \leq 1$, it vanishes on the true boundary, and

$$W \leq Bq^{-2}\phi \qquad (r > 5000). \tag{14}$$

To obtain a positive solution below W, consider the cube $Q = (-100, 100)^{21}$. Its closure lies strictly inside Ω: on it $r < 2000$ and $|y| \leq 100$, whereas $\ell \geq 2000 + \pi/2$. The function

$$\psi(z) = \prod_{i=1}^{21} \cos(\pi z_i/200) \quad (z \in Q)$$

satisfies $0 < \psi \leq 1$, zero boundary data and $-\Delta\psi = \lambda_Q \psi$, with $\lambda_Q = 21\pi^2/40000 < 1$. Its outward derivative is nonpositive on every face. Let $m_Q = \min_{\overline{Q}} W > 0$ and $\varepsilon = \min((1-\lambda_Q)/2, m_Q/2) > 0$. The zero extension of $\varepsilon\psi$ lies below W, and in Q

$$\mathcal{F}(\varepsilon\psi) = \varepsilon\psi[(1 - \lambda_Q) - \varepsilon\psi] \geq 0.$$

Formula (4), applied to the faces, proves its global weak subsolution inequality. Lemma 2 now gives a positive equilibrium T with $\varepsilon\psi \leq T \leq W \leq 1$ and zero Dirichlet data. For $r > 5000$ and $q > B$, (12) and (14) give

$$U(x, 0) \geq q^{-1} > Bq^{-2} \geq T(x, 0).$$

Such points exist because B is fixed and finite. The two equilibria are therefore distinct. Their values are in $[0, 1]$, so they solve both the polynomial Fisher–KPP equation and (1) with the global reaction (2). This proves Theorem 1.

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